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8. Construct a C program to simulate Round Robin scheduling algorithm with C.

Code:

```
main.c
1 #include<stdio.h>
2 int main(){
3     int bt[10],rem[10],wt[10],tat[10];
4     int n,tq,i,time=0,done;
5     printf("Enter number of processes: ");
6     scanf("%d",&n);
7     for(i=0;i<n;i++){
8         printf("Enter Burst Time: ");
9         scanf("%d",&bt[i]);
10        rem[i]=bt[i];
11    }
12    printf("Enter Time Quantum: ");
13    scanf("%d",&tq);
14    do{
15        done=1;
16        for(i=0;i<n;i++){
17            if(rem[i]>0){
18                done=0;
19                if(rem[i]>tq){
20                    time+=tq;
21                    rem[i]-=tq;
22                }
23                else{
24                    time+=rem[i];
25                    wt[i]=time-bt[i];
26                    rem[i]=0;
27                }
28            }
29        }
30    }while(done==0);
31    printf("\nProcess\tWT\tTAT\n");
32    for(i=0;i<n;i++){
33        tat[i]=bt[i]+wt[i];
34        printf("P%d\t%d\t%d\n",i+1,wt[i],tat[i]);
35    }
36    return 0;
37 }
```

Output:

```
Output
Enter number of processes: 3
Enter Burst Time: 10
Enter Burst Time: 5
Enter Burst Time: 8
Enter Time Quantum: 2

Process WT  TAT
P1  13  23
P2  10  15
P3  13  21

=== Code Execution Successful ===
```